Natural experiment

Hill of Towie, 21 turbines
1 retrofitted with vortex generators (Sep 2021),
20 unmodified for 9 months

open 10-min SCADA
reference-cell estimator
placebo turbines and
pseudo-retrofit dates

Retrofit effect

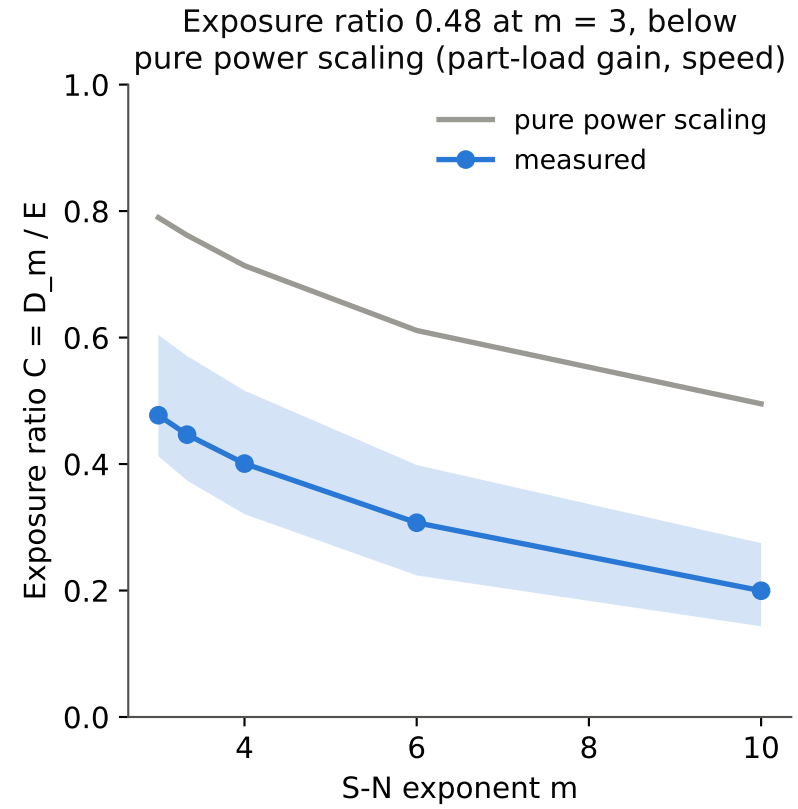
Energy

+4.8%

Time at rated

+1.7 pp

Torque exposure (m=3)

+2.3%

rank 1 of 21 turbines